Table 1: Prominent software suites for data streaming and analytics

Software suite	URL
KNIME Analytics Platform	*https://www.knime.com/knime-analytics-platform*
RapidMiner	*https://rapidminer.com/*
Apache Spark	*https://spark.apache.org/*
Pentaho	*https://www.hitachivantara.com/en-us/products/lumada-dataops/data-integration-analytics/download-pentaho.html*
Kafka Streams	*https://kafka.apache.org/*
Atlas.ti	*https://atlasti.com/*
Apache Storm	*https://storm.apache.org/*
Apache Hadoop	*https://hadoop.apache.org/releases.html*
Apache Cassandra	*http://cassandra.apache.org/download/*
Apache Flink	*https://flink.apache.org/*
OpenRefine	*https://openrefine.org/*
Data Cleaner	*https://github.com/datacleaner*
Apache Nifi	*https://nifi.apache.org/*

Assuring consistent data quality and allowing rapid analysis for business insights can be achieved by merging data from several silos into a single source of truth. Also, information from the point of sale (PoS) system can be processed in real-time using a streaming data pipeline. The stream processing engine may loop the pipeline's outputs back to the PoS system or send them to other applications like data repositories, marketing apps, and customer relationship management (CRM) software.

Each data pipeline has three main parts: a data source, some sort of processing step(s), and a final destination. A data pipeline's last stop may be referred to as a 'sink' in this context. Data pipelines facilitate the transfer of information between several systems, such as an application and a data warehouse, a data lake and an analytics database, or a data lake and a payment processing system. It is possible for a data pipeline to have a single source and a single destination, in which case the pipeline's sole purpose would be to effect changes to the data collection. A data pipeline exists wherever information is sent from point A to point B (or B to C to D).

More and more data is being moved between applications, and as organisations seek to build applications with small code bases that serve a very specific purpose (called 'microservices'), the efficiency of data pipelines has become crucial. Numerous data pipelines may be fed by information produced by a single system or application, and each data pipeline may in turn feed multiple other pipelines or applications.

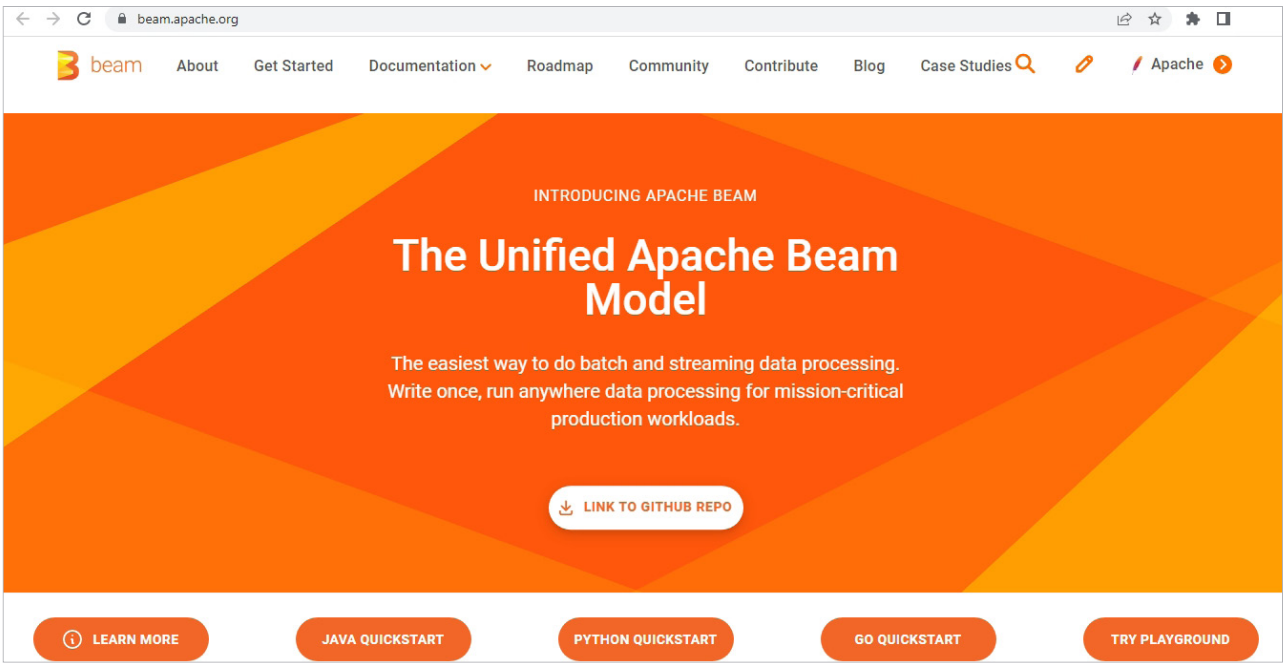

Figure 2: Official portal of Apache Beam